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#KubeCon #CloudNativeCon

I Tested 7 So You Only Need 1

Your First Gateway API Migration in 5 Minutes

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Keep the flow. Swap the middle



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※ Same way in. Same app out. I replace **only** the layer in the middle.

Every implementation passes the Core



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NGINX
Gateway Fabric



kgateway



Kong



traefik
proxy



Core · 7 the mandatory baseline, passed by every one

✓ host-routing

✓ path-routing

✓ header-routing

✓ tls-termination

✓ canary

✓ header-modifier

✓ cross-namespace



Who am I ?



<https://github.com/SysNet4Admin>



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2024

2025

2026



AAIF
AMBASSADOR



<https://www.linkedin.com/in/hoonjo/>



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Stage#1:

- Ingress NGINX Retired



Ingress NGINX Retired, not Ingress Deprecated



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Ingress NGINX Controller

2026.03

project archived



source: kubernetes.io

RETIRED

The community project
kubernetes/ingress-nginx

No more releases or patches
(**CVE-2025-1974** · **CVSS 9.8**)



ALIVE

The Ingress API Other controllers

Gateway API is the path forward



Four reasons this is worth doing



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Role-oriented

split by who owns what

infra · cluster · app

Expressive

fields, not annotations

header match · weighted traffic

Extensible

attach policies

without forking the core API

Portable

one spec

same YAML, many engines

Gateway API Introduction

gateway-api.sigs.k8s.io/docs/introduction/

Kubernetes docs · Gateway API

kubernetes.io/docs/concepts/services-networking/gateway/

Role-oriented: who owns what



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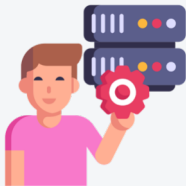
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GatewayClass



Infrastructure provider

which implementation to run



Gateway



Cluster operator

entry point, IP, TLS, listeners



HTTPRoute



App developer

my app's routes



Role-oriented: Ingress vs Gateway API



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INGRESS one object, one owner

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: jp-front
spec:
  ingressClassName: nginx # infra choice
  rules:
  - host: demo.kubecon.jp
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: jp-front
            port: {number: 80}
```

GATEWAY API split by role

```
# Gateway (cluster operator)
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata: {name: jp-gateway}
spec:
  gatewayClassName: nginx
  listeners:
  - {name: http, protocol: HTTP, port: 80}
---
# HTTPRoute (app developer)
kind: HTTPRoute
metadata: {name: jp-front}
spec:
  parentRefs: [{name: jp-gateway}]
  hostnames: [demo.kubecon.jp]
  rules:
  - backendRefs: [{name: jp-front, port: 80}]
```



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Stage#2:

- Which Gateway API?



Gateway API has 20+ implementations



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23 listed. I tested the 7 highlighted, under identical conditions.

 NGINX NGF	 Envoy Gateway	 Istio	 Cilium	 kgateway	 Kong (KGO)
 Traefik	 Contour	 Gloo Gateway	 GKE	 Google CSM	 Amazon EKS
 AWS LBC	 Azure AGC	 HAProxy	 Linkerd	 Agentgateway	 Kong (KIC)
 Alibaba CSM	 Airlock MGW	 Calico	 Kubenvornor	 Sunbeam	

Source: gateway-api.sigs.k8s.io/implementations (now adds Calico, Sunbeam) · PoC measured the 7 at v1.4

Core: all 7 are conformant



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Every implementation passes the 7 mandatory Core features. **Same baseline for all.**

		host	path	header	tls term	canary	header mod	Cross ns
	EG	✓	✓	✓	✓	✓	✓	✓
	Istio	✓	✓	✓	✓	✓	✓	✓
	Cilium	✓	✓	✓	✓	✓	✓	✓
	kgateway	✓	✓	✓	✓	✓	✓	✓
	NGF	✓	✓	✓	✓	✓	✓	✓
	Traefik	✓	✓	✓	✓	✓	✓	✓
	KGO	✓	✓	✓	✓	✓	✓	✓

Core 7 = routing (host/path/header), TLS termination, canary, header-modifier, cross-namespace. Snapshot v1.4 (2026.06)

Extended: common, differentiating



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Common (6) all ✓ | **differentiating 7** | Ext /13

		Common (6) all ✓	https-redir	timeout	backend-tls	gRPC	req-mirror	be-req-hdr	path-redir	Ext /13
	EG	6 ✓	✓	✓	✓	✓	✓	✓	✓	13 /13
	Istio	6 ✓	✓	✓	✓	✓	✓	✓	✓	13 /13
	Cilium	6 ✓	✓	✓	X	✓	✓	✓	✓	12 /13
	kgateway	6 ✓	✓	✓	X	✓	✓	✓	✓	12 /13
	NGF	6 ✓	✓	X	✓	✓	✓	.	✓	11 /13
	Traefik	6 ✓	✓	X	✓	✓	X	X	✓	10 /13
	KGO	6 ✓	X	X	X	X	.	.	X	6 /13

Common 6 = url-rewrite · response-header-mod · method/query-matching · websocket · listener-isolation. ✓ pass X fail · n/a · v1.4 snapshot

Which one fits you



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Widest feature breadth



Envoy Gateway, Istio

Smoothest from ingress-nginx



NGF

Already on Cilium CNI



Cilium

Going to a service mesh



Istio

Already on Traefik



Traefik

Envoy power without a full mesh



kgateway

Zero-downtime pod swaps



Istio, kgateway

Plugin ecosystem matters most



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Stage#3:

- ingress2gateway with{DEMO}



ingress2gateway converts your **Ingress**



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An official Gateway API tool. It reads your ingress-nginx Ingress and prints the Gateway API equivalent: a **Gateway** + **HTTPRoute**



```
$ ingress2gateway print -f ingress.yaml --providers ingress-nginx
```

```
-> Gateway + HTTPRoute (your routing, translated)
```

One command does the translation. The hard part is what it cannot translate

What converts, and what does not



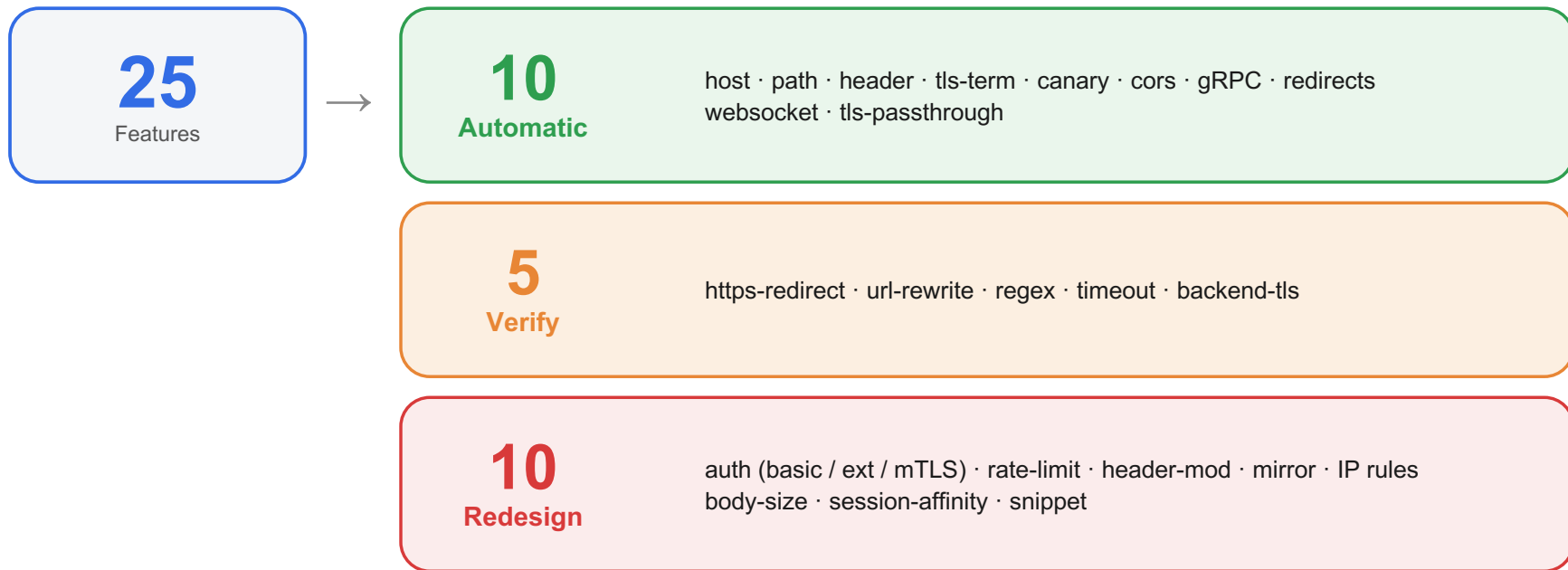
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25 ingress-nginx features, sorted by migration effort





Your first migration, in 3 stages

The Gateway is pre-staged and warm, so the live migration is a single cutover



The dot is the left-pane **curl**: it never stops. Green = serving, red = the one brief break.

LIVE DEMO

Stage 1 Working

ingress-nginx serves demo.kubecon.jp on 192.168.1.12

```
$ kubectl get ingress jp-front
```



curl: OK

LIVE DEMO

Stage 2 Broken (cut over)

Retire ingress-nginx; the pre-staged warm Gateway takes the same IP (~1s)

```
$ kubectl -n ingress-nginx delete svc ingress-nginx-controller
```



curl: OK -> FAIL -> OK

LIVE DEMO

Stage 3 Recovered

Same IP, same URL, now served by Gateway API

```
$ kubectl get gateway jp-gateway -n nginx-gateway -o wide # ADDRESS  
192.168.1.12
```



curl: OK



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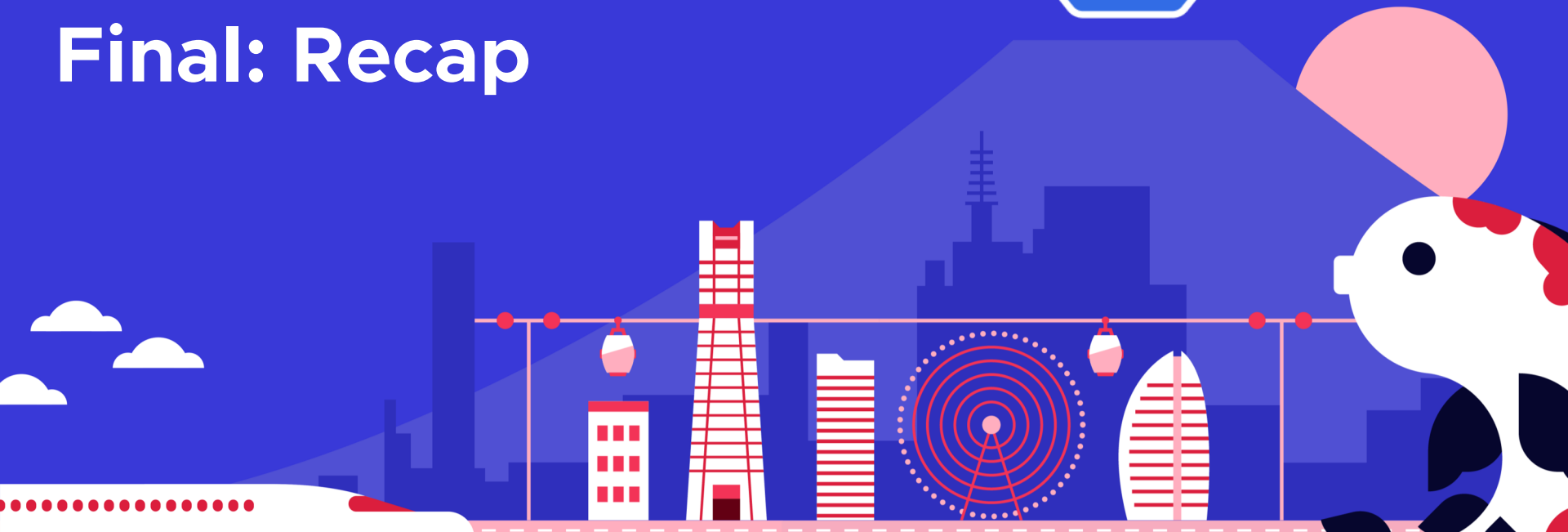


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Final: Recap



Why this matters now



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The clock ran out

ingress-nginx maintenance ended March 2026. No more patches

The ecosystem already moved

23 implementations, all Core-conformant. Ingress has no successor of its own

The next layer builds on it

AI gateways and inference routing extend Gateway API, not Ingress

Migrating now puts you on the platform the next layer is built on

Thank you. Questions?

Scan to try it yourself, everything is public.



PoC repository

github.com/sysnet4admin/Research



The 7-way comparison

kuberneteslab.dev/.../gateway-api-poc-v1.4



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<https://github.com/SysNet4Admin>



<https://www.linkedin.com/in/hoonjo/>